

Loyola Marymount University

CMSI 5350/EECE 5150 Machine Learning

Professor Lanyu Shang

Announcements

- Homework 3
  - Due this Friday 10/16
- PostCommit quiz
  - Gradient ascent!
- HW2 is graded
  - Brightspace: weighted grade
  - Gradescope: detailed comments



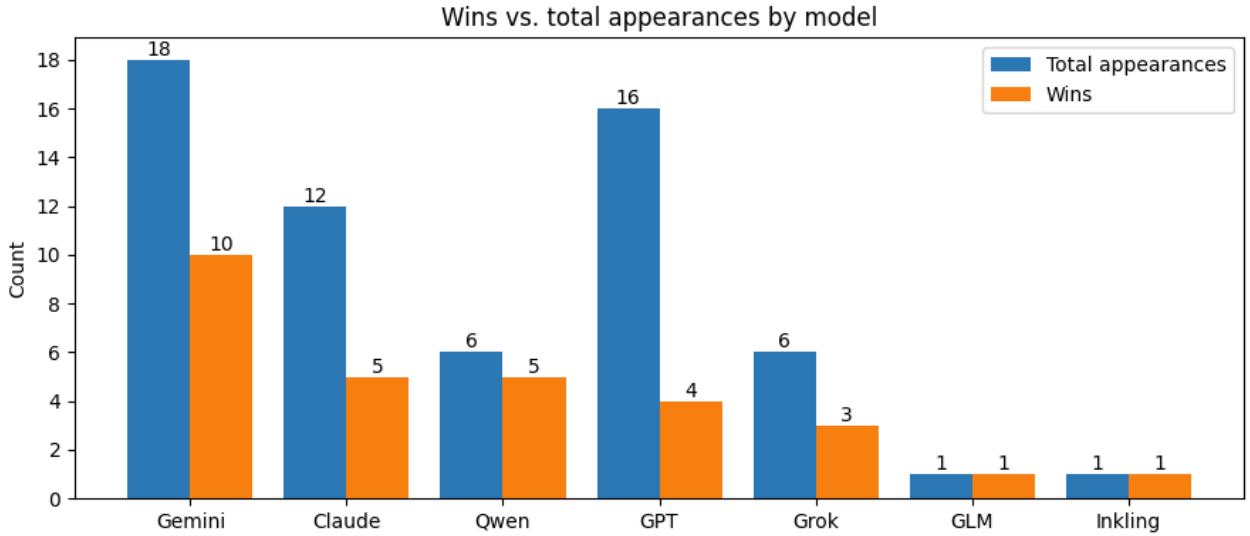
PCQ – HW1



PCQ – HW2

Homework 2 - GenAI

- Which one was the most/least popular?

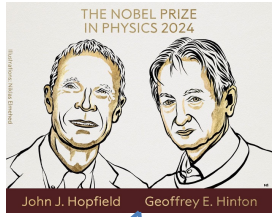


- Try more challenging questions in HW3?

Recent news

AMD × World Labs

A.M. TURING AWARD





National Medals of Science and
National Medals of Technology and Innovation, 2026



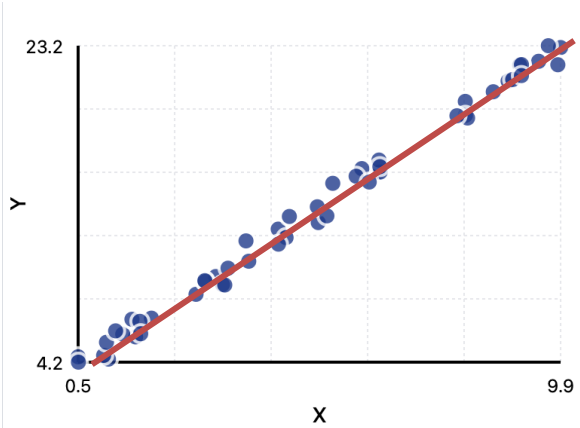
Queen Elizabeth Prize for Engineering, 2025

This lecture

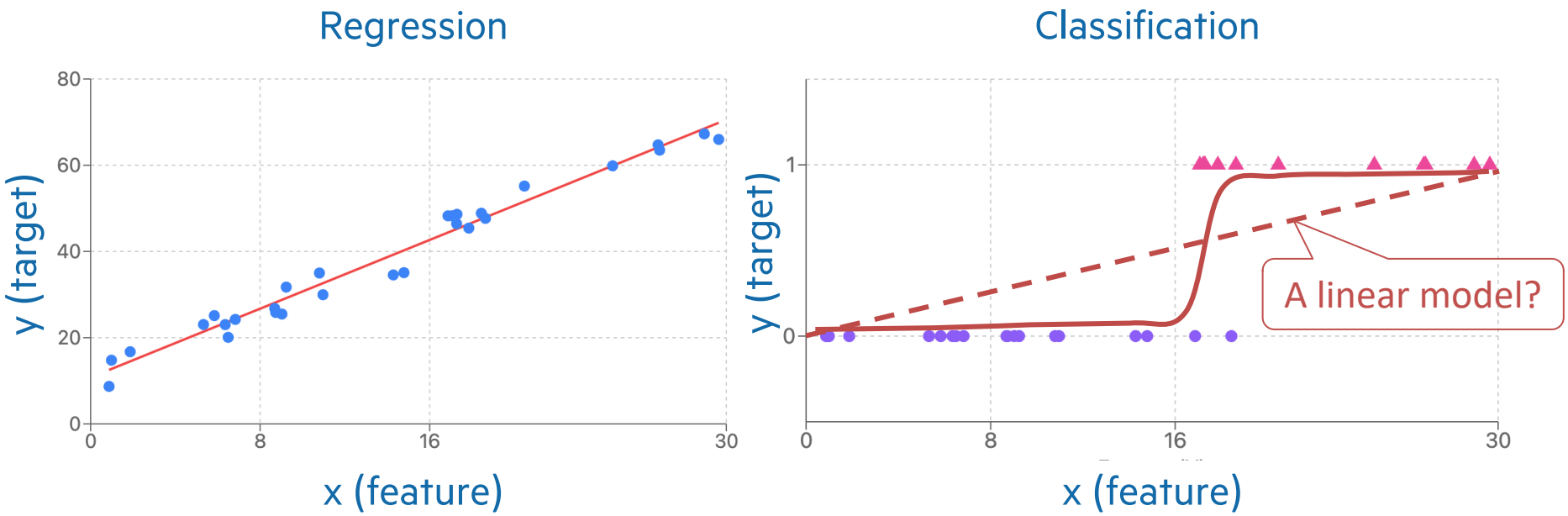
- Logistic regression
  - Problem setting
  - Objective function
  - Learning/training

Last lecture – linear regression

- Data $X = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \dots, \boldsymbol{x}^{(n)}\}$
  - Each $\boldsymbol{x}^{(i)} \in \mathbb{R}^d$ is a d-dimensional feature vector
- Labels $Y = \{y^{(1)}, y^{(2)}, \dots, y^{(n)}\}$
  - Each $y^{(i)} \in \mathbb{R}$
- Hypothesis set $\mathcal{H} = \{h \mid h: X \to Y\}$
  - $h_\theta(\boldsymbol{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d$
- Loss function
  - MSE: $J(\boldsymbol{\theta}) = \frac{1}{2n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$

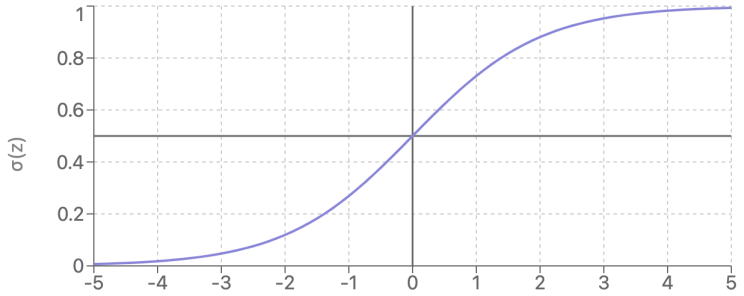


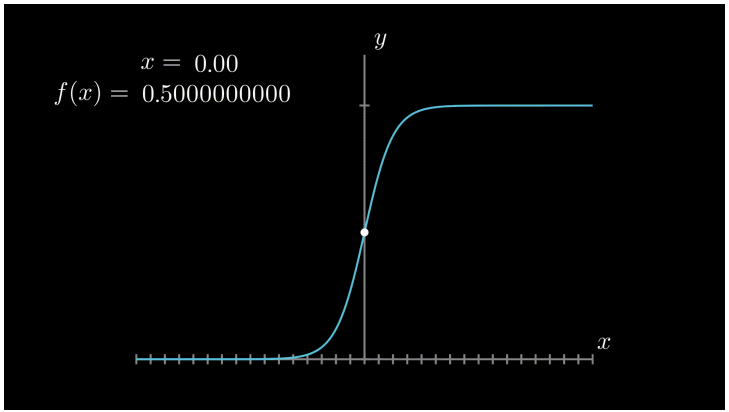
How about if the target is categorical?



Logistic function

- Logistic/sigmoid function
  - $\sigma(z) = \frac{1}{1+e^{-z}}$
  - $\sigma(z)$: $\mathbb{R} \rightarrow (0,1)$
  - $\sigma(0) = 0.5$
  - $\sigma(\infty) = 1$
  - $\sigma(-\infty) = 0$





Logistic regression – problem settings

- Data $X = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \dots, \boldsymbol{x}^{(n)}\}$
  - Each $\boldsymbol{x}^{(i)} \in \mathbb{R}^d$ is a d-dimensional feature vector
- Labels $Y = \{y^{(1)}, y^{(2)}, \dots, y^{(n)}\}$
  - Each $y^{(i)} \in \{0,1\}$ (categorical)
- Hypothesis set $\mathcal{H} = \{h \mid h: X \to Y\}$
  - Each hypothesis h is a probability function
  - $h_\theta(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T \boldsymbol{x}) = p(y = 1 \mid \boldsymbol{x}; \boldsymbol{\theta})$
- Loss function?

Yes! Logistic regression is a **classification** algorithm!

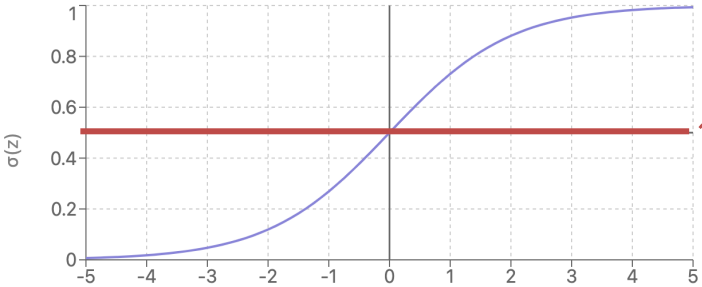
Logistic regression

- $h_\theta(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T \boldsymbol{x})$

$$= \frac{1}{1+e^{-\boldsymbol{\theta}^T \boldsymbol{x}}}$$

- Prediction/Decision rule
  - $\hat{y} = 1$ if $h_\theta(\boldsymbol{x}) \geq \boldsymbol{\tau}$
  - $\hat{y} = 0$ if $h_\theta(\boldsymbol{x}) < \boldsymbol{\tau}$
  - $\boldsymbol{\tau}$ is the threshold (decision boundary)
    - $\boldsymbol{\tau} = 0.5$ is the default threshold

Sigmoid function: $\sigma(z) = \frac{1}{1+e^{-z}}$



$\boldsymbol{\theta}^T \boldsymbol{x} \geq \mathbf{0} \leftrightarrow h_\theta(\boldsymbol{x}) \geq \boldsymbol{\tau} \leftrightarrow \hat{y} = 1$

$\boldsymbol{\theta}^T \boldsymbol{x} < \mathbf{0} \leftrightarrow h_\theta(\boldsymbol{x}) < \boldsymbol{\tau} \leftrightarrow \hat{y} = 0$

Sigmoid function: $\sigma(z) = \frac{1}{1+e^{-z}}$

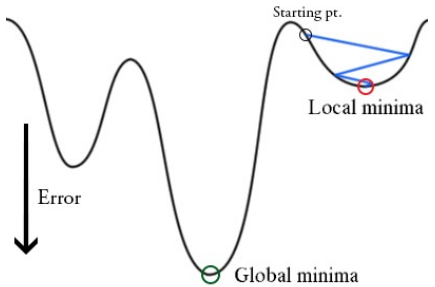
Example

- Credit card approval
  - $h_\theta(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T \boldsymbol{x})$
  - $\boldsymbol{\theta} = [-13, 0.02]^T$
  - $\boldsymbol{x} = [1, x_\text{score}]^T$
  - Given an applicant whose credit score is **680**, what is the probability their application will be **approved**?
  - The bank decides they want to be more **conservative**, what are the possible approaches?

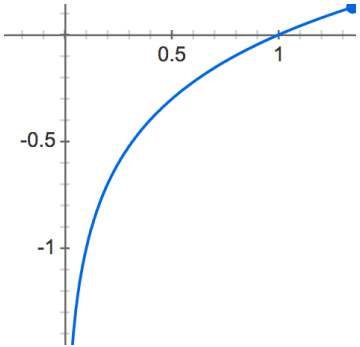
Solving logistic regression

- Loss/objective function
  - Linear regression?
    - $J(\boldsymbol{\theta}) = \frac{1}{2n}\sum_{i=1}^{n}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 = \frac{1}{2n}\sum_{i=1}^{n}(h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) - y^{(i)})^2$
    - Can we use mean squared loss to find optimal $\boldsymbol{\theta}$?
      - No!
      - Square of a σ function is **non-convex**
      - Gradient is weak/small -> slow learning

$$\sigma(\boldsymbol{\theta}^T\boldsymbol{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T\boldsymbol{x}^{(i)}}}$$



Loss/objective function



- Cross-entropy – taking the log
  - Binary labels
    - $\text{cost}(y, \hat{y}) = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$
    - Or $$\text{cost}\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}), y\right) = \begin{cases} -\log h_{\boldsymbol{\theta}}(\boldsymbol{x}), & \text{if } y = 1 \\ -\log\left(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x})\right), & \text{if } y = 0 \end{cases}$$
    - If $y = 1$,
      - when $\hat{y} \to 1, \log(\hat{y}) \to \log(1) = 0$ – No penalization!
      - when $\hat{y} \to 0, \log(\hat{y}) \to \log(0) = -\infty$ – Huge penalization!
    - If $y = 0$, same rules apply.

Loss/objective function

- Cross-entropy loss
  - $J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}[y \cdot \log(\hat{y}^{(i)}) + (1-y) \cdot \log(1-\hat{y}^{(i)})]$
  - Measures the "distance" between two distributions (how well $\hat{y}$ approximates y)
- Entropy? $H(X) = -\sum_{i=1}^{n} P(X=i)\log_2 P(X=i)$
  - $H(y) = -(P(y=0) \cdot \log P(y=0) + P(y=1) \cdot \log P(y=1))$
  - Measures **uncertainty/randomness** in a **single** distribution of y

Learning a logistic regression model

- Minimize objective function $J(\boldsymbol{\theta})$. How?
- Gradient descent!
  - $\theta_j \leftarrow \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\boldsymbol{\theta})$ (each dimension)
  - $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha \nabla J(\boldsymbol{\theta})$ (all dimension)
  - $\frac{\partial}{\partial \theta_j} J(\boldsymbol{\theta})$? -> worksheet

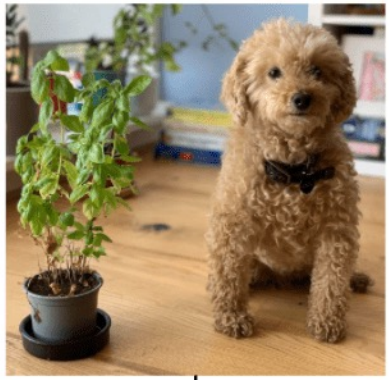
Multi-class classification

Binary Classification



Dog 0.9 | Not Dog 0.1

Multiclass Classification



Dog 0.5 | Cat 0.09 | Bus 0.01 | Plant 0.4



Multilabel Classification



Dog 0.8 | Cat 0.2 | Bus 0.04 | Plant 0.7

Multi-class logistic regression

$$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+\frac{1}{e^{z}}} = \frac{e^{z}}{e^{z}+1}$$

- Binary logistic regression

$$\hat{y} = h_\theta(\boldsymbol{x}) = \sigma(\boldsymbol{\theta}^T\boldsymbol{x}) = \frac{e^{\boldsymbol{\theta}^T\boldsymbol{x}}}{e^{\boldsymbol{\theta}^T\boldsymbol{x}} + \mathbf{1}}$$

$p(y = 1|\boldsymbol{x};\boldsymbol{\theta})$

Weight assigned to "y=1"

Weight assigned to "y=0"

- Multiclass logistic regression

Softmax function

$$p(y = k|\boldsymbol{x};\boldsymbol{\theta_1},\boldsymbol{\theta_2},\dots,\boldsymbol{\theta_K}) = h_{\boldsymbol{\theta_k}}(\boldsymbol{x}) = \frac{e^{\boldsymbol{\theta_k}^T\boldsymbol{x}}}{\sum_{j=1}^{K} e^{\boldsymbol{\theta_j}^T\boldsymbol{x}}}$$

Weight assigned to "y=j"

- Decision rule: $\hat{y} = \underset{k}{\operatorname{argmax}}\, h_{\boldsymbol{\theta_k}}(\boldsymbol{x})$

The **class** with the highest probability.